What Do We Mean by MIS?

Start with the system you can see. Then expand to the management and organization around it.

- Yesterday's Zhuiyi Technology case was more useful for competition and strategy than for isolating the role of MIS.
- MIS is a broad concept: technology + management + organization
- From an **end-user perspective**, MIS often appears as: a system, an application, a website
- That is a good starting point, but for this MBA course we also need the **manager perspective**: (1) how the system is managed (2) how it supports processes (3) how it is embedded in the organization
- Classic enterprise applications such as ERP, CRM, and SCM are still useful, but today we will use digital transformation to see the bigger picture
- A practical example: at WKU, MyWKU, KeanWise, and Canvas can be viewed as separate MIS from the end-user side, while managers also need to consider how they connect to processes, decisions, and organizational work

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MIS and Digital Transformation (DT)

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Learning Objectives

By the end of this session, you should be able to:

- Understand **why** firms pursue digital transformation
- Understand what is digital transformation
- Evaluate key trade-offs: efficiency vs. agility, own vs. partner, centralize vs. empower
- Explain how *ERP*, *SCM*, and *CRM* can support – but not guarantee – transformation
- Diagnose **why** digital transformation initiatives underperform

Kick Off

In your experience, what most often triggers digital transformation?

A. Cost and efficiency pressure

B. Customer experience / new demand

C. Competitive threat

D. Regulation / risk / crisis

Why DT

?? Tigger by → Technology alone
?? Tigger by → Old operating model fails

Pressure 1:
customer expectations

- Customers benchmark against the best digital experience they've had — anywhere.
- They demand speed, visibility, personalization, and seamless journeys.
- A great product no longer excuses a poor experience.

Pressure 2:
operating complexity

- Products, channels, suppliers, and data sources are more interconnected than before.
- Manual coordination breaks down as scale and variety increase.
- Most firms don't lack data – they lack usable, connected, decision-ready data.

Pressure 3:
competitive logic

- Rivals replicate products faster than ever.
- The battleground shifts to speed, integration, and ecosystem position.
- Transformation is defensive for some – a growth engine for others.

Three Familiar Transformation Triggers

Example A: bank

- Customers can open accounts online but still must visit a branch to complete key steps.
- The app exists, but the process is not truly digital.
- *Trans need:* redesign journey, compliance workflow, and decision rules – not just launch another app.

Example B: manufacturer

- Demand changes quickly, but production and procurement react too slowly.
- Forecasts, inventory, and logistics are fragmented across teams.
- *Trans need:* integrated planning, supply visibility, and shorter response cycles.

Example C: hospital/public service

- A digital front end exists, but professionals still rely on phone calls, spreadsheets, and workarounds.
- Users see a digital service; employees experience manual rework.
- *Trans need:* back-end integration, workflow redesign, and role clarity.

What Is DT

What it is

- A **strategic** and **organizational** redesign enabled by digital technologies.
- It changes how a firm creates value, operates, makes decisions, and serves customers.
- It often affects processes, roles, metrics, structure, and culture – not just systems.

What it is not

- Not simply buying new software.
- Not a website, app, dashboard, or AI pilot by itself.
- Not an IT project that leaves business processes and incentives unchanged.

What Is DT

Digitization – Converting analog information into a digital format; the foundation layer.

Digitalization — Leveraging digital data to streamline workflows, reduce manual effort, and improve efficiency.

Digital transformation — Rethinking strategy, structure, and value creation around digital — changing how the business competes and grows.

What Changes in DT

Customer interface

Channels
journeys
personalization
service speed

Operations

planning
workflow
coordination
cycle time

Decision model

data visibility
analytics
real-time signals
exceptions

Business model

pricing
channel logic
customization
ecosystem

Organization

roles
incentives
governance
skills

Potential Misread of DT

Mistake 1: technology-first

- "Let's buy the tool first and figure out use cases later."
- *Result:* expensive pilots, weak adoption, unclear value.

Mistake 2: front-end only

- The app or portal improves, but data, workflow, and incentives stay old.
- *Result:* customers see digital; employees still do manual workarounds.

Mistake 3: efficiency only

- The firm optimizes cost and control but ignores speed, flexibility, and new revenue logic.
- *Result:* better today, weaker tomorrow.

Digital transformation is difficult because it is easy to modernize the visible layer while leaving the **underlying business logic untouched**.

Three Managerial Tensions in DT

1. Efficiency vs agility

How much should the firm standardize for control and cost, and how much flexibility should it preserve for speed and customization?

2. Own vs outsource/partner

What should the firm build as an internal capability, and what should it source from vendors or ecosystem partners?

3. Centralize vs empower

Which decisions and data standards must be centralized, and where should business units or front-line teams retain autonomy?

Transformation is not only a technology problem. It is a set of operating model choices under uncertainty.

Tension 1 – Efficiency vs. Agility

Efficiency logic

- Standardize processes and data definitions
- Reduce waste and errors
- Increase visibility, consistency, and control
- Works well when demand is stable and variation is costly

Agility logic

- Respond faster to changing demand
- Support small batches, customization, and experimentation
- Allow faster local decisions and shorter loops
- Matters when customers, channels, or products change quickly

Good transformation does not choose one side forever. It decides where standardization creates value and where flexibility creates value.

Tension 2 – Own vs. Outsource/Partner

Why firms outsource or partner

- Faster time to market
- Access to missing technology or talent
- Lower upfront investment
- Easier scaling in well-understood domains

Why firms keep capabilities in-house

- Need tighter fit with strategy and data
- Need faster adaptation to changing market needs
- Need stronger control over customer experience or sensitive processes
- Want to build a long-term capability, not only buy a solution

The strategic question is not “build or buy?” It is “which capabilities must **remain distinctive**, and which can be **modularized**?”

Tension 3 – Centralize vs. Empower

Why centralize?

- One data standard
- Shared platforms and lower duplication
- Clear governance and accountability
- Better enterprise-wide visibility

Why empower local units?

- Closer to customers and operations
- Faster adaptation to local needs
- More ownership and experimentation
- Better fit for diverse business lines

Managerial challenge

- Centralize what must be common.
- Empower where speed and contextual knowledge matter.
- Without this balance, the firm becomes either chaotic or slow.

Many digital transformations fail not because the technology is wrong, but because governance is wrong.

Experience Exchange

From the end-user perspective

- Where do you see a “digital” service that still feels slow, fragmented, or inconsistent?
- What part of the experience is standardized well? What part still feels rigid?

From the manager / process-owner perspective

- Where does your company struggle to balance control and responsiveness?
- Which processes should stay standardized? Which should become more flexible?

Be concrete: what process, what user, what trade-off?

How to Conduct DT

1. Diagnose

- strategic pain points
- customer friction
- operational bottlenecks
- data gaps

2. Redesign

- customer journey
- operating model
- roles and decisions

3. Build foundations

- data
- platforms
- process integration
- standards

4. Scale adoption

- change management
- metrics
- governance
- capabilities

5. Learn

- iterate
- rebalance
- upgrade

Transformation is not a one-time implementation.
It is a staged redesign and learning process.

ERP, SCM, CRM

ERP

- Integrates core internal processes and data
- Useful for standardization, visibility, and planning discipline
- Strong backbone – but rarely the transformation by itself

SCM

- Connects planning, procurement, production, logistics, and suppliers
- Important when responsiveness depends on end-to-end coordination
- Especially critical in manufacturing and distribution-heavy settings

CRM

- Unifies customer information and touchpoints
- Useful for service, personalization, and commercial effectiveness
- More central in service-intensive or relationship-driven industries

AI in ERP

Oracle AI Agent Studio for Fusion Cloud Applications: Demo

Capability	Description
Digital assistant	AI chatbots handle customer and employee inquiries, resolving issues without human intervention.
Predictive analytics	Anticipates equipment failures and recommends preventive maintenance.
Anomaly detection	Flags irregularities in financial data entries and transactions.
Improve product development & innovation	Surfaces new product ideas and optimizes features and assortment.
Financial forecasting	Models financial scenarios based on economic, policy, or market shifts.
Optimize workflows	Supports planning, inventory, logistics, and inspection across the supply chain.

AI in SCM

- Demand forecasting
  - Machine learning, predictive analytics
- Inventory management
  - AI-powered robotics, IOT sensors
- Logistics and transportation optimization
  - Route optimization algorithms, real-time tracking
- Supplier relationship management
  - AI-based supplier risk assessment, chatbots
- Quality control and predictive maintenance
  - Computer vision, AI sensors

AI-Driven Supply Chain Transformation (Heineken at AIPCon 6)

AI in CRM

[3 Ways Salesforce Uses AI with Data and CRM]

Capability	Description
Digital assistant agent	AI chatbots handle customer inquiries, resolve issues, and capture details without human intervention.
Field service routing optimization	Uses AI and ML to optimize scheduling and resource allocation, helping teams complete more jobs per day.
Lead conversion and sales opportunity win probability	Scores leads and opportunities by win likelihood; flags deals needing attention.
Pricing optimization	Recommends pricing and discounts to maximize revenue while maintaining close probability.
Predictive scoring	Predicts which leads are most likely to engage and convert.

Different Industries Transform Differently

Manufacturing / supply chain heavy

- Pain points: planning, inventory, downtime, quality, coordination
- Typical focus: SCM, smart operations, forecasting, traceability
- Trade-off: scale efficiency vs customization

Service / relationship intensive

- Pain points: handoffs, inconsistent service, weak visibility into the customer
- Typical focus: CRM, workflow redesign, omnichannel experience
- Trade-off: personalization vs process consistency

Public / regulated sectors

- Pain points: compliance, legacy systems, fragmented ownership
- Typical focus: process integration, governance, transparency
- Trade-off: control and accountability vs speed and innovation

The right transformation logic depends on the operating bottleneck and the source of value in the industry.

Why DT Fail

1. Wrong problem

- Tool-first mindset
- No clear value pool
- Weak link to strategy

2. Weak redesign

- Old workflows survive
- Data stays fragmented
- Responsibilities stay unclear

3. Weak adoption

- Little frontline buy-in
- Poor incentives
- Insufficient training and change management

4. Weak governance

- Too centralized or too fragmented
- No sequencing
- No metrics & learning loop

Most transformation failures are not caused by software defects. They come from poor diagnosis, weak redesign, weak adoption, and weak governance.

When AI Enters DT

What AI can add

- Better forecasting and anomaly detection
- Smarter service and recommendation
- Automation of repetitive decisions
- Faster experimentation and pattern recognition

What AI does not solve

- Broken data foundations
- Poor process design
- Unclear incentives or governance
- Lack of trust and adoption

Practical note

- AI can accelerate digital transformation, but it usually creates value only when the underlying operating model and data foundation already make sense.

Case – Tsingtao Brewery

1) What are the major challenges for Tsingtao Brewery before launching its digital strategy? Discuss why Tsingtao Brewery chose Blue Ocean Strategy to solve these challenges with the help of digital transformation.

2) How can we assess Tsingtao Brewery's financial performance after the execution of its digital transformation strategy?

3) Compare the business model before and after Tsingtao Brewery started its digital transformation strategy. What other resources and capabilities could also be strategically essential?

4) What activities will make Tsingtao Brewery turn its digital strategy into results?

5) Please discuss any potential difficulties that Tsingtao Brewery could face during its digital transformation.